Cyclistic Bike-Share Analysis – Case Study

1. Project Title

Cyclistic Bike-Share Analysis: Member vs Casual Rider Behavior

2. Objective

The objective of this analysis was to understand how annual members and casual riders use Cyclistic bikes differently and identify opportunities to convert casual riders into annual members.

3. Tools Used

• SQL (Google BigQuery)
• Tableau Public

4. Data Cleaning & Preparation

The dataset containing bike trip records was uploaded to Google BigQuery. Data cleaning steps included:
• Splitting and uploading large CSV files.
• Removing duplicate ride_id records using SQL window functions.
• Creating a ride_length column using timestamp difference between start and end times.
• Filtering invalid rides such as negative durations or rides longer than 24 hours.

5. Analysis & Key Insights

• Members take significantly more rides than casual riders.
• Casual riders have longer average ride durations (~21 minutes) compared to members (~12 minutes).
• Casual riders ride more on weekends, while members ride more frequently during weekdays.
• Ride activity increases during summer months, peaking around July and August.

6. Dashboard

Interactive Tableau Dashboard: Cyclistic Bike-Share Analysis Dashboard
https://public.tableau.com/views/CyclisticBike-ShareAnalysisMemberVsCasualBehaviour/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Business Recommendations

Based on the analysis, Cyclistic could implement the following strategies:

- Target casual riders with weekend promotions.
- Offer seasonal membership discounts during peak riding months.
- Highlight cost savings and convenience of annual memberships in marketing campaigns.